

General Science





About the Course

Learners are introduced to scientific activities and technology with a nod toward historical context. Specifically, students learn about plate tectonics, anatomy and medicine, and architectural engineering. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 6

About Science: Grade 6

In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level and time required are transitional as we gently prepare for the middle grades.

Science: Grade 6

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 6

In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level, historical context, and time required are transitional as we gently prepare for the middle grades.

General Science: Grade 6

For sixth-grade students or possibly hungry fifth-graders or seventh-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate

level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
6	General Science: Grade 6 2 times/week 30 min	Plate Tectonics The Art of Construction Phineas Gage

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 6				
Natural History: Grade 6		General Science: Grade 6 Nature Walks & Scouting: Grades 1-8	General Science: Grade 6 Nature Notebook: Grade 6	Labs: Grade 6



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 6

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 6

Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 6

- ☐ Term 1: any local geology site (e.g. a quarry or certain parks)
- ☐ Term 2: hospital open house
- ☐ Term 3: any buildings or bridges

Term Prep & Teacher Tips

Science: Grade 6

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term

and/or any that support special topics chosen by students:

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☐ Term 1: specific rock types p.748-759

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- less than 1/2 c white vinegar
- recycled cardboard, roughly 3' x 3'
- books or wood blocks to adjust height
- 3 colored markers
- 2 or more apples (Term 2)
- plate
- knife
- cutting board
- plastic wrap
- roughly 2' x 2' piece of substrate (recycled cardboard, wood scrap, foam core)
- 15 sheets of multipurpose paper
- any weights for testing bridge
- various supplies by student design
- printables
- computer
- optional: camera, such as found on most electronic devices
- optional: antiseptic treatments
- optional: plastic ruler
- optional: fabric scrap
- optional: wax paper
- optional: fan/hair dryer
- optional: needle and thread
- optional: cereal box to model a building
- optional: rocks
- optional: sand paper
- optional: 3-7 weights with a hole, e.g. washers or spools
- optional: string or strong thread
- optional: hammer and nail
- optional: block of scrap wood
- optional: rope
- optional: nutcracker
- optional: 4 raw eggs (Term 3)
- optional: 8 sturdy egg cups
- optional: 8 cotton balls
- optional: scrap plywood (might be the same substrate as above)
- optional: cylindrical balloon
- optional: cup that fits the top of balloon
- optional: books of various sizes



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

General Science: Grade 6



Plate Tectonics



The Art of Construction



Phineas Gage



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

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Alveary Grade 6 Science Kit



Household Items - General Science: Grade 6



Student Microscope



Prepared Microscope Slides



Quick Links

(No Quick Links Assigned)

[Click THIS text
or scan the QR
code for links.](#)



General Science: Grade 6

How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even

repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

- Notice if there were any dates that they want to keep for their Book of Centuries.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
- These can be viewed as alternative ways to engage.

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[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1

30m General Science: Grade 6 - Lesson 1

God Created

Materials: The Holy Bible

→ INTRO

You have been learning about scientific activities in our modern world. Like all knowledge, our culture affects these activities, and in return, what we learn affects our culture. This has been going on throughout human history. Early scientists primarily studied the Things they could see and assumed that they were always this way, but eventually, they began to make some curious observations that suggested that maybe Things hadn't always been this way. These observations were shocking and confusing to these men and women. This was a major discovery that led many naturalists to question all they thought they knew and understood. It led some to reject science; others to reject their faith in God; and still others to be awestruck at the elegance and majesty of the Creator. What were these discoveries all about, and how did they change the direction of science? These are some very big ideas that require a lot of time to think about. Before getting started with the science, let's spend some time with our Bibles.

→ READ, NARRATE, & DISCUSS

Genesis 1-2

- What is the same in Genesis 1 and 2, and what is different? How do these two chapters work together to tell us about God?

★ TEACHER NOTE

This lesson consists of reflection on the Christian story of creation. Teachers might choose to allow students to do/review this independently, to engage in the reading/discussion, or to skip this one.

WEEK 1

30m General Science: Grade 6 - Lesson 2

Making Sense of What We See

Materials: Plate Tectonics

→ INTRO

Some of the discoveries that made scientists think that maybe Things changed over time came from studying rocks. Look at the title and cover. Do you have ideas about plate tectonics? We'll begin the story today with some of those early scientists.

Notice the sidebars in the text, like the one on p.14-15. As you become more independent with your reading, it can be helpful to think about how to deal with these kinds of "rabbit trails," which can enhance the reading. Because it interrupts the main line of thought, it is usually best to finish the section and then come back to the sidebar.

→ READ & NARRATE

Ch.1 p.11-13, 16-17 "From some" - "to be connected."

→ READ, NARRATE, & DISCUSS

p.14-15 "Around the" - "region's geology."

WEEK 2

30m General Science: Grade 6 - Lesson 3

Seismic Disasters

Materials: Plate Tectonics

→ RECAP

Recall what we read in the last passage. Today we'll read about seismic activity in Italy and Japan.

→ READ & NARRATE

∞ Article Link: The Destruction of Pompeii

→ READ, NARRATE, & DISCUSS

p.17-20 "Pompeii" - "with the natural world."

WEEK 2

30m General Science: Grade 6 - Lesson 4

Renaissance Ideas

Materials: Plate Tectonics

PREP: Read Teacher Tip

→ RECAP

Tell me about the occurrences in Italy and Japan. Naturalists continued to observe for a long time, but then our story leaps forward during the Renaissance.

→ READ, NARRATE, & DISCUSS

p.20-27 "After the decline" - "centuries earlier."

- Nicolaus Steno's work presented, for the first time, the idea that the Earth and even life itself were dynamic. Have you ever learned something so new and surprising that it seemed to change everything? What did you do with this new information?
- While Steno did abandon his scientific pursuits when he entered religious life, many others then and now pursue scientific endeavors and religious vocations hand-in-hand. Daniello Bartoli, Paolo Casati, and Maria Gaetana Agnesi are just a few from Steno's time period. How many can you

★ TEACHER NOTE

Conflict between religion and Steno's work is alluded to on p.25: "Like many of" - "remainder of his life." Teachers might choose to allow it to pass by the way for now (this idea is a major theme in Grade 7-8), to engage in discussion, or to check out the book about Steno in Extra Helpings. The same conflict is mentioned in the supplemental video (1:16-1:31).

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They

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Term 1

find?

→ SUPPLEMENTAL

The author notes that ideas often inspire people who come later. One scientist you should know who is not mentioned in our book is Mary Anning:
∞ Video Link: Mary Anning (3:40)

are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 3

30m General Science: Grade 6 - Lesson 5

Rocks and Hutton's Theory

Materials: Plate Tectonics

→ RECAP

Tell me about what Renaissance thinkers brought to the story. Before we read about the theory of plate tectonics, look at the figure on p.28. You may continue to look back at it during the passage.

→ READ, NARRATE, & DISCUSS

p.29-35 "We now know" - "parts of the puzzle?"

- Discuss the main points of James Hutton's theory. Use words, pictures, or diagrams.

★ TEACHER NOTE

The age of the Earth is mentioned along with Hutton's Theory of the Earth on p.33-34: "He also claimed" - "wind or rain." Teachers might engage in discussion, present alternative theories, or check one of the books in Extra Helpings for additional discussion.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 3

30m General Science: Grade 6 - Lesson 6

Oceanography and Seismology

Materials: Plate Tectonics

→ RECAP

Recall the last reading. We've read about paleontology and geology. Today we have two more types of study that will reveal more puzzle pieces: oceanography and seismology.

→ READ & NARRATE

p.35-36 "Exploring the Oceans" - "sea level."

→ READ, NARRATE, & DISCUSS

p.36-41 "Finding ways" - "accurate measurement."

WEEK 4

30m General Science: Grade 6 - Lesson 7

Continental Drift

Materials: Plate Tectonics

→ RECAP

Recall the last passage. Today we begin to put the puzzle together.

→ READ, NARRATE, & DISCUSS

p.41, 43-46 "Continental Drift" - "and accepted."

- Discuss the main points of Alfred Wegener's theory. Use words, pictures, or diagrams.
- Do you think the idea of continental drift impacted the way that scientists organize and classify living things? How might it influence their understanding of which plants and animals are more closely related?

→ SUPPLEMENTAL

∞ Resource Link: NatGeo Continental Drift

→ NOTE

Read Abraham Ortelius p.42 for interest, if desired.

★ TEACHER NOTE

The age of the Earth is mentioned along with Wegener's theory of continental drift on p.43-45 and in the supplemental link: "Wegener believed" - "and Canada." Teachers might engage in discussion or consider alternative theories they would like to share.

WEEK 4

30m General Science: Grade 6 - Lesson 8

The Theory is Born

Materials: Plate Tectonics

→ RECAP

Recall the last reading. Today we'll learn a bit about Alfred Wegener.

→ READ, NARRATE, & DISCUSS

p.49-52 "The theory" - "moon and Mars."

WEEK 5

30m General Science: Grade 6 - Lesson 9

Convection and Plate Tectonics

Materials: Plate Tectonics

→ RECAP

Recall the last reading. Today we'll learn a bit about two more scientists.

→ READ & NARRATE

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Term 1

p.52-55, 58 "Dan McKenzie" - "discussed below." → READ, NARRATE, & DISCUSS p.56-57 "Jason Morgan" - "plate tectonics."	
WEEK 5 30m General Science: Grade 6 - Lesson 10 <i>Measuring Earthquakes</i> Materials: Plate Tectonics PREP: Read Teacher Tip → RECAP Recall the last reading. Our focus today is on the two scientists who developed the Richter scale, which is used to measure earthquake strength. → READ, NARRATE, & DISCUSS p.58-63 "Charles Richter" - "Uppsala in 1955."	★ TEACHER TIP Are your learner(s) demonstrating more familiarity/interest in volcanoes, earthquakes, or other aspects of plate tectonics? More interest/ease in discussing how scientists developed the theory over time? If not, consider discussing together more often, exploring extra helpings, or taking a field trip to a natural history museum. Reach out through Contact Us, if you need help.
WEEK 6 30m General Science: Grade 6 - Lesson 11 <i>Seafloor Spreading</i> Materials: Plate Tectonics → RECAP Recall the last passage. Today we read about a scientist who studied the ocean floor. → READ, NARRATE, & DISCUSS p.63-64 "Harry Hess" - "sister planets."	★ TEACHER NOTE Evolution is mentioned at the end of the reading on p.64. Teachers might allow it to pass by the way or consider how they would like to discuss. • NATURE NOTEBOOK Prompt for General Science in Quick Link.
WEEK 6 30m General Science: Grade 6 - Lesson 12 <i>The Earthquake Zone</i> Materials: Plate Tectonics PREP: Read Teacher Tip → RECAP Recall the last passage. Today we read about two scientists who studied the Ring of Fire. → READ, NARRATE, & DISCUSS p.64-68 "Kiyoo Wadati" - "in 1968." → SUPPLEMENTAL ∞ Video Link: Ring of Natural Disasters (6:34)	★ TEACHER TIP Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).
WEEK 7 30m General Science: Grade 6 - Lesson 13 <i>Geological Dating</i> Materials: Plate Tectonics → RECAP Recall the last passage. Today we read about our last scientist who developed a method of geological dating. → READ, NARRATE, & DISCUSS p.68-69 "Arthur Holmes" - "the earth." → SUPPLEMENTAL ∞ Video Link: How does radiocarbon dating work? (2:11) ∞ Video Link: Earth's Entire History (4:38)	★ TEACHER NOTE The age of the Earth is mentioned at the top of p.69 and in the second supplemental video. Isotope dating is introduced in the first supplemental video. Teachers might allow it to pass by the way, engage in discussion, consider alternative theories they would like to share, or skip the videos.
WEEK 7 30m General Science: Grade 6 - Lesson 14 <i>Discovering Puzzle Pieces</i> Materials: Plate Tectonics → RECAP Recall the last passage. Let's look at how all of this comes together in the theory of plate tectonics. → READ, NARRATE, & DISCUSS p.71-76 "In 1912" - "continental drift." • Scientists are currently debating whether spreading, like Harry Hess and Marie Tharp described, is a requirement for life on a planet and whether Earth is unique in this feature. Why would they theorize that seafloor spreading would be important for life? What kind of evidence do you suppose they might look for to support or refute such a theory?	★ TEACHER NOTE The timeline of Earth's magnetic reversal is mentioned on p.73: "every three hundred thousand to seven hundred thousand years." Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

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Term 1

WEEK 8	30m General Science: Grade 6 - Lesson 15 <i>How it Works</i> Materials: Plate Tectonics → RECAP Tell me what we read last time: how is this theory coming together so far? Let's read about the different movements that can occur according to the theory. → READ, NARRATE, & DISCUSS p.76-83 "The Breakthrough" - "opposite directions."	★ TEACHER NOTE The age of the Earth is mentioned on p.78 ("Some three hundred" - "change is continuous."), 79 ("but over thousands" - "amounts of movement."), and 81 ("Millions of years" - "ten million years."). Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.
WEEK 8	30m General Science: Grade 6 - Lesson 16 <i>Natural Disasters</i> Materials: Plate Tectonics → RECAP Tell me about the different types of plate movement according to plate tectonic theory. Today we read about the results of that movement: earthquakes, tsunamis, and volcanoes. → READ, NARRATE, & DISCUSS p.83-89 "Earthquakes" - "plate tectonics." Describe earthquakes and volcanoes in terms of plate tectonic theory. You may use words, pictures, or diagrams.	★ TEACHER NOTE The age of the Earth is mentioned on p.88: "Recent studies suggest" - "the earth's existence." Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.
WEEK 9	30m General Science: Grade 6 - Lesson 17 <i>Technology</i> Materials: Plate Tectonics → RECAP Recall what you have learned about plate tectonic theory. Have you read about computer modeling being used to study weather and ocean movement? Do you think that computers could help predict tectonic movement, as well? → READ, NARRATE, & DISCUSS p.90-91 "Advances" - "geodynamic models."	
WEEK 9	30m General Science: Grade 6 - Lesson 18 <i>Planning for Plate Tectonics</i> Materials: Plate Tectonics → RECAP Recall the last passage. Can you think of ways that people's lives are affected by this understanding of plate tectonics? → READ & NARRATE p.93-95, 98-100 "In previous" - "structural damage." → READ, NARRATE, & DISCUSS p.96-97 "To predict" - "to evacuate."	
WEEK 10	30m General Science: Grade 6 - Lesson 19 <i>Planetary Change</i> Materials: Plate Tectonics → RECAP Recall what you read in the last passage. Today's reading tells how plate tectonic theory affects how we study the planets. → READ, NARRATE, & DISCUSS p.100-105 "Plate Tectonics" - "might form."	★ TEACHER NOTE The timescale of Earth's existence is mentioned on p.105: "Such a process" - "ten million years." Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.
WEEK 10	30m General Science: Grade 6 - Lesson 20 <i>What Lies Ahead</i> Materials: Plate Tectonics → RECAP Recall what you read in the last passage. → READ, NARRATE, & DISCUSS p.106-109 "What Lies" - "years to come."	★ TEACHER NOTE The timescale of Earth's existence, past and present, is mentioned several times on p.106. Teachers might choose to allow it to pass by the way, to engage in discussion, or to consider alternative theories they would like to share.

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Term 1

→ SUPPLEMENTAL

∞ Video Link: Alfred Wegener (7:51)

WEEK 11

30m General Science: Grade 6 - Lesson 21

Catch Up

Materials: Plate Tectonics

PREP: Read Teacher Tip

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP

When student(s) take exams next week, evaluate what they recall, but more importantly, whether plate tectonics, its effects, or how scientists arrived at the theory are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.

WEEK 11

30m General Science: Grade 6 - Lesson 22

Catch Up

Materials: Plate Tectonics

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 12

30m General Science: Grade 6 - Lesson 23

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:
Plate Tectonics

WEEK 12

30m General Science: Grade 6 - Lesson 24

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:
Plate Tectonics

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[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 1

30m General Science: Grade 6 - Lesson 25

Horrible Accident in Vermont

Materials: Phineas Gage

PREP: Read Teacher Tip

→ INTRO

This is the story of a man who suffered a terrible head injury. Can you spot it in the picture on the cover?

→ READ, NARRATE, & DISCUSS

p.1-8 "The most unlucky" - "happened to him."

• Throughout this course, we'll be learning about anatomy, the study of the body's structures, and physiology, the study of how they work. We organize all of the different parts into systems, even though they really all function together. Why do you think scientists would study it in smaller portions like that? Do you know of any of these systems yet?

★ TEACHER TIP

Note that the next lesson is an activity day!

WEEK 1

30m General Science: Grade 6 - Lesson 26

Body Systems

Materials: video links, paper, scissors, pencils/markers

PREP: Read Teacher Tip

→ RECAP

Recall what happened to Phineas Gage. The skin is part of one system, while the brain is part of another. Do you know what systems they are each part of? Watch the video for an introduction to body systems. Listen for where the skin and brain belong. Were any other systems involved in Phineas's injury?

→ VIEW, NARRATE, & DISCUSS

∞ Video Link: Body Systems (3:33)

- Consider how incredible is the formation of our bodies. Consider how many intricate details must go right during the formation of each person. Every living being is truly a miracle!
- An idea that is beautifully displayed in our body systems is that "form follows function." How are the forms in our bodies related to the functions we need?

→ ACTIVITY

Let's represent the body systems by making a paper chain, something like in the video link. How many sides do you need to represent the systems? How will you represent each system?

∞ Video Link: Paper Chain People (3:16)

★ TEACHER NOTE

The reproductive system is mentioned in the lesson video (2:17-2:22). Teachers might allow it to pass by the way or decide how they would like to discuss further with students.

★ TEACHER TIP

Note that there is an optional activity next week using the microscope and prepared slides.

WEEK 2

30m General Science: Grade 6 - Lesson 27

Tending to Phineas

Materials: Phineas Gage, optional: microscope, prepared slides

PREP: Read Teacher Tip

→ RECAP

Recall the last passage. What happened to Phineas? What do you think will happen next?

→ READ, NARRATE, & DISCUSS

p.8-15 "That's how Dr." - "postsurgical infections."

- Is this what you expected to happen to Phineas? What most surprised you? What would it have been like if you were one of Phineas's friends on the scene?

→ SUPPLEMENTAL

If you have a microscope and prepared slides, this is a great time to compare some of the many different types of cells at work in the body. Compare the dog tissue slides and the blood smear. Describe the different forms/shapes that you see. Consider how the shapes might contribute to different functions.

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

★ TEACHER TIP

Note that the next lesson is an additional day to work on Lab 1!

WEEK 2

30m General Science: Grade 6 - Lesson 28

Lab: Classification

Materials: Grade 6 Lab Book and materials listed within

→ ACTIVITY

Use this time to continue work on Lab 1.

Term 2

WEEK 3

30m General Science: Grade 6 - Lesson 29

Wounds

Materials: video links

→ RECAP

Where did we leave Phineas? He's got this terrible wound right now. Let's find out more about what doctors knew and didn't know in 1848, and what needs to happen for his wounds to heal. Students don't need to remember the scientific terms in the video. Focus on the meaning of the ideas.

→ VIEW, NARRATE, & DISCUSS

∞ Video Link: Germ Theory (4:39)

- Did you notice how Koch learned about and tested his ideas about cholera? He used what scientists call an animal model, a mouse. The model stands in for a human so that scientists can learn more about how the human body works without harming a human. What do you think are the pros and cons of this? (e.g., good that humans aren't harmed, but not fair to animals; good if the model fits, but bad if we don't realize its limitations; etc.)
- If doctors didn't yet know about germs in 1848, how was Phineas going to heal?

→ VIEW, NARRATE, & DISCUSS

∞ Video Link: How Wounds Heal (4:01)

- Which body systems did you notice working together to heal the wound in the video?

WEEK 3

30m General Science: Grade 6 - Lesson 30

Phineas Lives

Materials: Phineas Gage

→ RECAP

Remind me where we left Phineas. Let's find out what happens next.

→ READ, NARRATE, & DISCUSS

p.15-22 "None of this" - "make arrangements."

- Dr. Harlow said, "I dressed him. God healed him." Why did he say this? What does this tell us about Dr. Harlow?
- Is Phineas healed? What does it mean to be healed? Is healing the same as being cured and exactly as you were before? What happened when Jesus healed? Is this the same or different from what is meant by Phineas's doctors?

WEEK 4

30m General Science: Grade 6 - Lesson 31

How We Know What We Know

Materials: video links

→ RECAP

Phineas Gage lives, but he still has a problem. Tell me about that. Part of his problem is that doctors know so little about how the brain works. Even for basic, everyday tasks, our bodies are performing an amazing dance. Let's watch the video for just one example.

→ VIEW, NARRATE, & DISCUSS

∞ Video Link: Oxygen's Surprisingly Complex Journey (5:10)

- How many systems were working together in the video? How do you think we figured all of that out? Let's find out how William Harvey figured out that blood circulates.

→ VIEW, NARRATE, & DISCUSS

∞ Video Link: Ceaseless Motion (to 4:54)

- Did you notice how William Harvey finally figured it out? (He used models and performed experiments.)
- Phineas is what we call a case study, a unique opportunity to learn through an example. Scientists will learn a lot about the brain by observing him, but they are limited. Discuss why they are limited. (e.g., they cannot experiment, they cannot understand the complexity of what has happened to him) How do you suppose models help to bring understanding?

WEEK 4

30m General Science: Grade 6 - Lesson 32

The Doctors

Materials: Phineas Gage, image link

PREP: Read Teacher Tip

→ RECAP

Phineas still has a problem, but there are a lot of doctors who are interested in learning more about his case. Let's do a little picture study to get an idea of what these doctors are like. What

★ TEACHER NOTE

The painting included for picture study is a graphic representation of surgery. Teachers should preview if they have concerns. They might choose to present the picture like any other, to direct students to the setting and doctors rather than the patient, or to skip this introduction.

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Term 2

do you notice in The Gross Clinic? ∞ Image Link: The Gross Clinic → READ, NARRATE, & DISCUSS p.23-27 "In the winter" - "wouldn't be human." • "...both groups are slightly right, but mostly wrong. Yet their wrong theories... will steer our knowledge of the brain in the right direction." How can someone be both right and wrong? Have you ever experienced this? How can a wrong idea help us get to the truth?	★ TEACHER TIP Note that there is an optional activity next week using the microscope and prepared slides.
WEEK 5 30m General Science: Grade 6 - Lesson 33 <i>The Brain</i> 📄 Materials: Phineas Gage PREP: Read Teacher Tip → RECAP Recall the last reading. → READ, NARRATE, & DISCUSS p.27-34 "The cortex is" - "acts, and reacts." • If you have a microscope and prepared slides, check out the pig motor neuron. How does this compare to the other cells you observed previously? How is its form similar/different from the others? How is its function similar/different based on what you have read?	★ TEACHER TIP Note that the next 3 lessons are a clay modeling activity!
WEEK 5 30m General Science: Grade 6 - Lesson 34 <i>Clay Modeling</i> 📄 Materials: clay or other modeling material, toothpicks, knives, or any tools you like for modeling, optional: paint PREP: Read Teacher Tip → INTRO In the last reading, you read about the form of the brain. Using what you know and the illustrations in your book, create a clay model of the brain over the next 3 lessons. → ACTIVITY	★ TEACHER TIP Are your learner(s) demonstrating more familiarity/interest in the human body or how it works? More interest/ease in discussing how scientists developed their understanding? If not, consider discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a human body exhibit. Reach out through Contact Us, if you need help.
WEEK 6 30m General Science: Grade 6 - Lesson 35 <i>Clay Modeling</i> 📄 Materials: clay or other modeling material, toothpicks, knives, or any tools you like for modeling, optional: paint → ACTIVITY Continue your model of the brain today.	
WEEK 6 30m General Science: Grade 6 - Lesson 36 <i>Clay Modeling</i> 📄 Materials: clay or other modeling material, toothpicks, knives, or any tools you like for modeling, optional: paint → ACTIVITY Complete your model of the brain today.	
WEEK 7 30m General Science: Grade 6 - Lesson 37 <i>The Brain Debate</i> 📄 Materials: Phineas Gage PREP: Read Teacher Tip → RECAP Recap the last reading. → READ, NARRATE, & DISCUSS p.34-42 "The Boston Doctors" - "out of town." • Discuss the arguments of the Whole-Brainers and the Localizers. Use an outline, if helpful. → SUPPLEMENTAL ∞ Video Link: The Great Brain Debate (5:20)	★ TEACHER TIP Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

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Term 2

WEEK 7

30m General Science: Grade 6 - Lesson 38

Following Phineas

Materials: Phineas Gage

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.43-49 "The story of" - "everywhere Phineas goes."

WEEK 8

30m General Science: Grade 6 - Lesson 39

The end of Phineas

Materials: Phineas Gage

→ RECAP

Recall the last reading.

→ READ, NARRATE, & DISCUSS

p.49-57 "In 1859, Phineas" - "shipped to Massachusetts?"

• Let's return to this question of whether Phineas is healed. Does he seem healed? Is he trying to heal? What do you think when you hear about his life?

WEEK 8

30m General Science: Grade 6 - Lesson 40

Doctors Again

Materials: Phineas Gage

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.57-64 "What a request." - "stop acting human?"

WEEK 9

30m General Science: Grade 6 - Lesson 41

Putting Phineas Together Again

Materials: Phineas Gage

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.65-69 "In our time" - "down to one."

→ SUPPLEMENTAL

∞ Article & Image Link: Smithsonian Magazine

WEEK 9

30m General Science: Grade 6 - Lesson 42

Putting Phineas Together Again

Materials: Phineas Gage

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.69-75 "Brainvox plots it" - "Gage drove fast."

• What makes us human? Is it any of the ideas that the author suggests? Or is it something else? If someone loses these abilities through birth, injury, or illness, do they cease to be human? If your faith tradition provides guidance, what does it say?

★ TEACHER NOTE

The question of what makes us human is introduced on p.70: "Humans have" - "an iron rod." Teachers might choose to allow it to pass by the way or to discuss the philosophical/theological implications.

WEEK 10

30m General Science: Grade 6 - Lesson 43

An Update on Phineas in Chile

Materials: article link

PREP: Read Teacher Tip

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

★ TEACHER TIP

Note that the next lesson is an additional day to work on the final lab project!

General Science: Grade 6

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Term 2

∞ Article Link: The Curious Case of Phineas Gage • Does this portrayal of Phineas in Chile affect your ideas about his life? Why or why not?	
WEEK 10 30m General Science: Grade 6 - Lesson 44 <i>Lab: Student Design</i> 📄 Materials: Grade 6 Lab Book and materials listed within → ACTIVITY Use this time to continue work on the final model project. → SUPPLEMENTAL A fun video to watch while you think about your amazing body! ∞ Video Link: How Old is Your Body? (3:09)	
WEEK 11 30m General Science: Grade 6 - Lesson 45 <i>Catch Up</i> 📄 Materials: Phineas Gage PREP: Read Teacher Tip → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When student(s) take exams next week, evaluate what they recall, but more importantly, whether anatomy, physiology, or how scientists arrived at their understanding are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.
WEEK 11 30m General Science: Grade 6 - Lesson 46 <i>Catch Up</i> 📄 Materials: Phineas Gage → CATCH UP Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12 30m General Science: Grade 6 - Lesson 47 <i>Exam Week</i> → EXAMS Answer question(s) related to course. Questions will come from: Phineas Gage	
WEEK 12 30m General Science: Grade 6 - Lesson 48 <i>Exam Week</i> → EXAMS Answer question(s) related to course. Questions will come from: Phineas Gage	

General Science: Grade 6

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Term 3

WEEK 1

30m General Science: Grade 6 - Lesson 49

From Cave to Skyscraper

Materials: Art of Construction

PREP: Read Teacher Tip

→ INTRO

Look at the title and cover. In this book, you will learn how man has used the forces of nature to develop knowledge in structural design. From primitive tents to modern pneumatic roofs, architecture may be one of man's oldest scientific pursuits. Note that you will need your text for lab this term because the lab instruction comes from the text.

→ READ, NARRATE, & DISCUSS

p.1-8 "Thirty thousand" - "and pins."

→ SUPPLEMENTAL

∞ Image Link: Former Sears Tower

∞ Image Link: Burj Khalifa (tallest building)

★ TEACHER NOTE

Reading begins with a mention of the age of the Earth: "Thirty thousand years..." Teachers might choose to allow it to pass by the way, decide how to discuss, or consider alternative theories they would like to present.

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 1

30m General Science: Grade 6 - Lesson 50

Building a Tent

Materials: Art of Construction, lab notebook

→ RECAP

Recap the last reading. Today is day 1 (the Introduction) of the Building a Tent lab, so that students will be ready for a full day of activity on lab day.

→ READ, NARRATE, & DISCUSS

p.9-10 "The purpose of" - "makes it stand up."

→ LAB NOTEBOOK

In your lab notebook, compose a few sentences to answer the introductory questions, "What do I know about building a tent?" and "What do I plan to do?" If you have extra time, copy some of the diagrams from your book.

WEEK 2

30m General Science: Grade 6 - Lesson 51

Building a Tent

Materials: Art of Construction, lab notebook

PREP: Read Teacher Tip

→ RECAP

Recap the last reading. Today is day 3 (the Analysis and Conclusions) of the Building a Tent lab, so students will continue their reading.

→ READ, NARRATE, & DISCUSS

p.11-15 "The straw" - "from spreading apart."

→ LAB NOTEBOOK

In your lab notebook, compose a few sentences to answer the questions, "What did I learn?" and "What would I do differently if I built this structure again?" You may want to include a picture or diagram of your tent.

★ TEACHER TIP

Three readings are scheduled this week rather than having an activity during lab. Teachers wanting to continue Lab 1 or do a supplemental activity during the lab time will need to adjust the schedule or have students catch up on the reading at another time.

WEEK 2

30m General Science: Grade 6 - Lesson 52

What is a Beam?

Materials: Art of Construction

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.17-20 "Some elements" - "in the beams."

- Describe tension and compression using words, drawings, and/or diagrams.

→ NOTE

The next reading (p.21-26 "Since all" - "bronze rods.") occurs in this week's lab, so be sure to read it before continuing to the next lesson.

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Term 3

WEEK 3

30m General Science: Grade 6 - Lesson 53

Behavior of Materials

Materials: Art of Construction

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.26-31 "To get" - "collapsing entirely."

→ SUPPLEMENTAL

The Tacoma Narrows Bridge was nicknamed "Galloping Gertie" by construction workers. The rhythmic movement increased over time until its collapse (as seen in the video link) just 4 months later. The footage demonstrates the impressive strength and elasticity of the materials, even though they could not withstand the forces of nature.

∞ Video Link: Tacoma Narrows Bridge (2:35)

WEEK 3

30m General Science: Grade 6 - Lesson 54

The Floor of Your Room

Materials: Art of Construction

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.33-37 "If the earth" - "design engineer."

- Try calculating the total load on the floor of your room!

→ NOTE

The next reading (p.39-41 "The best way" - "near the column axis.") occurs in this week's lab, so be sure to read it before continuing to the next lesson.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 4

30m General Science: Grade 6 - Lesson 55

Constructing for Strength

Materials: Art of Construction, ruler

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.41-46 "Since I-columns" - "(Figure 6.14)."

WEEK 4

30m General Science: Grade 6 - Lesson 56

A Steel Frame... Made Out of Paper

Materials: Art of Construction

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.47-48 "In your paper" - "and their connections."

Be sure that you understand the structure you will build. Use your lesson time today to think about how you will use the columns/beams previously made. How will you assemble your structure? If helpful, you may want to outline or diagram your plans in your lab notebook, so you have a clear idea for the next lesson.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 5

30m General Science: Grade 6 - Lesson 57

The Foundation

Materials: Art of Construction

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.51-56 "The construction" - "going on."

General Science: Grade 6

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Term 3

WEEK 5

30m General Science: Grade 6 - Lesson 58

Forces that Unbalance

Materials: Art of Construction, optional: book, cereal box, sandpaper, a few rocks, fan/hairdryer, plastic ruler

PREP: Read Teacher Tip

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.57-63 "The purpose" - "the ruler."

- Explain equilibrium using words, drawings, and/or diagrams.

★ TEACHER TIP

Are your learner(s) demonstrating more familiarity/interest in structural design or engineering? More interest/ease in discussing the art of engineering? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a science museum with engineering or building exhibits. Reach out through Contact Us, if you need help.

WEEK 6

30m General Science: Grade 6 - Lesson 59

Wind Drift

Materials: Art of Construction, optional: fan/hairdryer, needle, thread, cereal box, weight (e.g., spool or metal washer)

PREP: Read Teacher Tip

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.65-69 "The horizontal" - "weight substantially."

- Calculate the force of wind on the building where you live or school as described on p.65. What would be the wind force of a 50 mph wind? A 100 mph wind?

→ SUPPLEMENTAL

- ∞ Video Link: John Hancock Tower (1:17)
- ∞ Video Link: John Hancock Tower Update (1:06)

★ TEACHER TIP

There are many constructions in the book that we don't have time for in the lesson schedule, but students can try any of these based on interest and as time permits!

WEEK 6

30m General Science: Grade 6 - Lesson 60

Earthquakes

Materials: Art of Construction, optional: sandpaper, cereal box, hammer, nail, block of scrap wood, rope

PREP: Read Teacher Tip

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.70-74 "The tuned" - "people died."

→ SUPPLEMENTAL

- ∞ Video Link: Swaying Skyscrapers (4:15)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 7

30m General Science: Grade 6 - Lesson 61

Using Strings to Carry Load

Materials: Art of Construction, optional: thread/string, 3-7 paper clips, 3-7 weights (e.g., spools or metal washers)

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.75-80 "Wires and" - "from it."

WEEK 7

30m General Science: Grade 6 - Lesson 62

Suspension Bridges

Materials: Art of Construction, optional: wooden ruler, thread/string, weight

→ RECAP

Recap the last reading.

→ READ, NARRATE, & DISCUSS

p.80-83 "The adaptability" - "suspension bridges."

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Term 3

→ SUPPLEMENTAL ∞ Image Link: Akashi-Kaikyo Bridge	
WEEK 8 30m General Science: Grade 6 - Lesson 63 <i>From Sticks and Stones to Bridges and Buttresses</i> Materials: Art of Construction, optional: thread/string, paper clip, weight, nutcracker → RECAP Recap the last reading. → READ, NARRATE, & DISCUSS p.85-94 "We have" - "them vertically." → SUPPLEMENTAL ∞ Image Link: Flying Buttress ∞ Image Link: New River Gorge Bridge	• NATURE NOTEBOOK Prompt for General Science in Outdoor Work Quick Link.
WEEK 8 30m General Science: Grade 6 - Lesson 64 <i>From Strings and Sticks to Trusses</i> Materials: Art of Construction, optional: thread/string, nutcracker, weight → RECAP Recap the last reading. → READ, NARRATE, & DISCUSS p.95-98 "If useful" - "(Figure 12.7)." → NOTE The next reading (p.99-103 "To build a" - "efficient trusses are.") occurs in this week's lab, so be sure to read it before continuing to the next lesson.	
WEEK 9 30m General Science: Grade 6 - Lesson 65 <i>Using Trusses</i> Materials: Art of Construction → RECAP Recap the last reading. → READ, NARRATE, & DISCUSS p.103-108 "Figure 12.14" - "out of trusses."	
WEEK 9 30m General Science: Grade 6 - Lesson 66 <i>Shape and Strength</i> Materials: Art of Construction, optional: paper, pencil, 3-5 books PREP: Read Teacher Tip → RECAP Recap the last reading. → READ, NARRATE, & DISCUSS p.109-118 "If you" - "its material."	★ TEACHER TIP There is an optional activity next week that requires 4 raw eggs.
WEEK 10 30m General Science: Grade 6 - Lesson 67 <i>Getting Creative and Complex</i> Materials: Art of Construction → RECAP Recap the last reading. → READ, NARRATE, & DISCUSS p.119-123 "The simple" - "with the roof."	
WEEK 10 30m General Science: Grade 6 - Lesson 68 <i>Creative and Complex</i> Materials: Art of Construction, optional: 4 raw eggs, 8 egg cups, 8 cotton balls, plywood scrap, scrap fabric or bandana → RECAP Recap the last reading.	

General Science: Grade 6

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Term 3

→ **READ, NARRATE, & DISCUSS**
p.123-133 "Domes have" - "four columns."

WEEK 11 30m General Science: Grade 6 - Lesson 69

Air as a Construction Material?

Materials: Art of Construction, optional: cylindrical balloon, cup that fits balloon, small books, thread/string, weight (e.g., spool or metal washer)

PREP: Read Teacher Tip

→ **RECAP**
Recap the last reading.

→ **READ, NARRATE, & DISCUSS**
p.135-143 "If you" - "man lives."

★ TEACHER TIP

When student(s) take exams next week, evaluate what they recall, but more importantly, whether the design, engineering, or art of structures are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help or have any concerns about readiness for the next stage in relationship building.

WEEK 11 30m General Science: Grade 6 - Lesson 70

Catch Up

Materials: Art of Construction

→ **CATCH UP**
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 12 30m General Science: Grade 6 - Lesson 71

Exam Week

→ **EXAMS**
Answer question(s) related to course.

Questions will come from:
Art of Construction

WEEK 12 30m General Science: Grade 6 - Lesson 72

Exam Week

→ **EXAMS**
Answer question(s) related to course.

Questions will come from:
Art of Construction

General Science: Grade 6

Examination

Term 1

- Plate Tectonics: Draw or diagram 3 types of plate movement and what they cause. OR Explain what is meant by the terms continental drift and seafloor spreading.
- Explain how you did one investigation in science lab this term and what you learned from it.

Term 2

- Phineas Gage: Explain why animals are used as models in science. OR Tell how the life of Phineas Gage has contributed to our understanding of the brain.
- Explain how you did one investigation in science lab this term and what you learned from it.

Term 3

- The Art of Construction: Explain how a structure's form comes from its needed function. OR The author describes architecture as both art and engineering. Give 3 examples that demonstrate this idea.
- Explain how you did one construction this term and what you learned from it.